

KESTREL — autonomous drone security analyst

One docked drone: ~57,600 frames per shift against a free tier of ~40 model requests per minute. Every decision below follows from that ratio.

MEASURED ON THIS MACHINE, NOT QUOTED

YOLO11s 12-15 ms · VLM 1.28 s · deep tier 57-84 s

gate skips 82.5% · 8/8 scenarios · 100 tests · P@k 0.975

INGEST — ONE INTERFACE, FOUR SOURCES

Video file

6 CC BY 4.0 clips · 768×432 to 1920×1080

Uploaded footage

an operator's own clip plus a typed location

Scripted frames

text descriptions, per the brief's literal spec

Telemetry simulator

position, altitude, gimbal, battery, wind, light

PERCEPTION — FIVE TIERS, EACH EARNS THE NEXT

TIER 0 · Gate

Perceptual hash · pixel delta · embedding novelty — all on CPU

free

skips 82.5%

The gate protects model spend, not GPU time. Local detection costs no API budget, so it runs on every sampled frame; the gate governs only the hosted tiers.

TIER 1 · Detect

YOLO11s (closed-set COCO) on-device
Grounding DINO when the query is open

12-15 ms

no API

TIER 1.5 · Track

ByteTrack, frame-rate aware
identity that survives occlusion

~free

30 s buffer

TIER 2 · Project

pixel → world via telemetry homography
→ named zone + honest accuracy radius

<1 ms

flat ground

TIER 2.5 · Embed

joint image/text, 2048-d, one space
re-identification and appearance search

cheap

NIM

TIER 3 · Perceive

VLM returns a structured SceneGraph:
objects, colour, kind, activity, anomalies
never prose, because prose is unqueryable

1.28 s

gated only

TIER 4 · Escalate

90B deep re-look, asynchronous
upgrades the record after the fact
routed by the kind of doubt

57-84 s

off critical path

MEMORY — WHY A PATTERN IS VISIBLE AT ALL

Entities

appearance embedding + attribute agreement +
spatio-temporal plausibility → one subject across days

"7th visit"

Temporal pyramid

L0 frame → L1 clip → L2 event → L3 shift → L4 day
salience-weighted, so the unusual survives compression
8 h ≈ 1.7 M tokens → ~12 k queryable

140:1

Normalcy baseline

counts per (zone, hour, class), z-scored
abstains below three days of history
"first vehicle at the dock at 03:00 in 14 days"

abstains

Hybrid retrieval

structured SQL ⊕ caption vectors ⊕ image vectors,
fused by Reciprocal Rank Fusion
declares when a retriever could not run

P@k 0.975

REASONING — DECLARATIVE, TEMPORAL, BACKTESTED

Rule engine — 12 condition types

dwelt · sequence · count_in_window · absence_of_person
baseline_anomaly · zone · time window · label · confidence

English → rule compiler

validated against the site's real zone list, then
backtested over indexed history before it may fire

Triage

contradicting evidence suppresses · mitigating evidence scales

ACTION — THE CLOSED LOOP

Dispatchable alert

coordinates ± accuracy · bearing · ETA from dock
recommended altitude · geofence verdict

Mission planner

waypoints checked against battery reserve,
wind, altitude ceiling and daylight

Human approval — a separate endpoint

the agent may propose; only a person may execute

CONTROL PLANE — CONVERSATION OVER EVERY CAPABILITY

Ask KESTREL

route → plan → tool loop → verify → respond
27 tools · 8 classes · 4 gated behind approval
answers render as live UI, not paragraphs

Scope guard

off-domain questions refused locally, before any tool runs

Citation check

every cited id must resolve to a real tool result

the loop closes: a closer vantage re-enters perception

proposal only, never execution

FOUNDATIONS — CROSS-CUTTING

Storage

SQLite + sqlite-vec, numpy fallback
frames, detections, entities, memory, alerts
skipped frames persisted too — the skip rate is the claim

Audit ledger

hash(n) = SHA-256(hash(n-1) || record(n))
every alert, rule change and approval appended
altering history invalidates every hash after it

Providers

Groq primary · NVIDIA NIM failover
circuit breaker · rate limiter · cost meter
cassette record/replay — runs with no API key at all

Surfaces

console · 3D globe · site map · investigate
entities · rules · evals · architecture
simulated data labelled everywhere it appears

LEGEND

tier-0 gate (CPU, free)

on-device (GPU, no API budget)

hosted model (rate-limited, metered)

human decision required

primary data path

feedback / secondary

enforcement boundary

Latencies are measured on this machine, not quoted from vendors. Telemetry is simulated: there is no aircraft. Only the flagship site carries real footage; every other site is labelled SIMULATED wherever it appears.